Gradient-Projected MUDVI: Combining Volume-Aware Memory Selection with Constrained Updates and Relationship-Shift Detection for Continual EEG Decoding

Abstract

Online continual decoding of EEG signals from sequentially arriving subjects suffers from catastrophic forgetting: as the decoder adapts to new subjects, its performance on previously learned subjects degrades. Duan et al. proposed MUDVI, a memory-based approach that keeps a balanced, informative buffer across subjects and jointly trains on buffer samples and incoming data. While effective, MUDVI's update step is a *soft* mechanism – nothing guarantees that a training step on new data will not increase the loss on memorized subjects, and its subject-shift detector monitors only the marginal distribution of extracted features, which can miss shifts in the underlying signal-to-label relationship. We propose two additions, adapted from a continual-learning framework originally built for regression on physiological time series (EM-ReSeleCT), and rewritten here for the classification setting native to EEG decoding: (1) a magnitude-aware gradient projection step that gives a hard, per-update guarantee against memory-loss degradation, and (2) a confidence-based change-point detector that flags subject shifts by watching a frozen reference model's loss stream, catching relationship shifts that a marginal-distribution detector cannot see. Both additions are designed as drop-in modules that plug into MUDVI's existing cluster-based memory buffer and shift-detection pipeline without altering its volume-imbalance handling, which remains the correct solution to a problem the source regression framework does not address.

1 Motivation

MUDVI's replay mechanism trains jointly on the incoming batch and a sampled subset of the memory buffer, relying on the *combined* gradient to keep old-subject performance intact. This is a statistical hope, not a guarantee: if the sampled memory batch is small, unrepresentative, or the new-subject gradient is large relative to it, a single update can still erase previously learned decoding behavior. This is consistent with Duan et al.'s own results – their reported backward transfer (BWT) for MUDVI is still negative in several settings (e.g. -6.52 on BCI IV-2a imbalanced, -8.26 balanced), indicating measurable forgetting persists even with the proposed buffer.

Separately, MUDVI's subject-shift detector (kernel-based MMD test on a projected distance metric) only tracks whether the *distribution of extracted features* has shifted. It has no mechanism for detecting a shift in the *relationship* between signal and decoded label – for instance gradual electrode drift, attentional fatigue, or a subtle new-subject pattern that does not immediately move the feature distribution but does change what the correct decoding boundary should be. MUDVI's own reported shift-detection accuracy (72.5% BCI IV-2a, 63.2% DEAP, 83.6% SEED) leaves room for a complementary detector.

A related continual-learning framework, EM-ReSeleCT, was developed for a regression task (glucose-insulin dynamics on the OhioT1DM dataset) and addresses both weaknesses directly: a magnitude-aware gradient projection rule gives a hard non-degradation constraint on every training step, and a residual-based detector catches shifts in the input–output relationship rather than just the input distribution. Both mechanisms are reformulated below for classification and integrated into MUDVI's existing pipeline.

2 Background: The Two Source Methods

2.1 MUDVI (Duan et al.)

MUDVI keeps a hierarchical memory buffer with one cluster per detected subject. For each cluster C, an importance score $\mathcal{I}_C$ is computed from the gradient norm of its members, and the probability of a sample being replaced is

$$\mathcal{A}_C \propto -\left(1 - \frac{n_C}{n}\right)\mathcal{I}_C, \qquad \mathcal{P}_t^C = \frac{e^{\mathcal{A}_C}}{\sum_{k=1}^{L_t-1} e^{\mathcal{A}_k}} \tag{1}$$

For incoming data x^t, the probability of moving it into memory is

$$P_t^{in} = \frac{e^{\mathcal{J}_t}}{\lambda e^{\mathcal{I}_\mathcal{M}} + e^{\mathcal{J}_t}} \tag{2}$$

Subject identity, when unknown, is detected via a kernel two-sample (MMD) test on a distance metric d_t computed between the current feature vector and a moving average of past features. This detector operates purely on the marginal distribution of extracted features and does not reference model predictions or loss.

Training proceeds by sampling from memory and jointly minimizing loss on the sampled memory and the incoming batch – a standard experience-replay style update, with no explicit constraint on the resulting gradient direction.

2.2 EM-ReSeleCT (regression formulation)

EM-ReSeleCT was built for streaming physiological regression, where a model predicts a continuous target $y_i \in \mathbb{R}^q$ from input $x_i \in \mathbb{R}^p$, with squared-error risk $R(\theta, D) = \frac{1}{|D|}\sum \|y_i - f(x_i;\theta)\|^2$. Two mechanisms are relevant here.

Constrained update. At every arriving segment, a memory gradient g_k^* and an incoming-data gradient g_{k+1} are computed. If they point in conflicting directions ($\langle g_k^*, g_{k+1}\rangle < 0$), the smaller-magnitude gradient is projected onto the larger one so the resulting update cannot increase the risk measured on memory:

$$\tilde{g} = \begin{cases} g_{k+1} - \dfrac{\langle g_{k+1}, g_k^*\rangle}{\|g_k^*\|_2^2}\, g_k^*, & \|g_{k+1}\| \geq \|g_k^*\| \\ g_k^* - \dfrac{\langle g_{k+1}, g_k^*\rangle}{\|g_{k+1}\|_2^2}\, g_{k+1}, & \|g_k^*\| \geq \|g_{k+1}\| \end{cases} \tag{3}$$

Relationship-shift detection. A model $\hat{\theta}_{D_1}$ is frozen after training on the first segment. The residual stream $r_t = y_t - f(x_t; \hat{\theta}_{D_1})$ is monitored with a change-point test (Narrowest-Over-Threshold, using a Gaussian likelihood ratio test between a one-line and two-line fit). A detected shift in r_t indicates the input–output relationship has changed, independent of whether the raw input distribution has moved.

Both mechanisms depend on numeric targets (squared-error gradients, numeric residuals), which do not exist in a classification setting and must be reformulated.

3 Proposed Method: Gradient-Projected MUDVI (GP-MUDVI)

3.1 Setting

Let a streaming EEG sample be (x_i, y_i), with $x_i \in \mathbb{R}^{t\times n}$ an EEG segment (temporal span t, n channels) and $y_i \in \{1, \ldots, K\}$ its class label. The decoder $f(\cdot;\theta)$ outputs class probabilities $\hat{p}_i \in \mathbb{R}^K$, with cross-entropy loss

$$\ell_i(\theta) = -\log \hat{p}_{i,y_i}, \qquad R(\theta, D) = \frac{1}{|D|} \sum_{i=1}^{|D|} \ell_i(\theta) \tag{4}$$

The memory buffer M retains MUDVI's cluster-per-subject structure and volume/informativeness-aware update rule unchanged.

3.2 Addition 1 – Constrained Replay via Gradient Projection

At each training step on incoming batch D_{k+1}, compute:

$$g_k^* = \frac{\partial}{\partial \theta} R(\theta, M)\Big|_{\theta=\hat{\theta}_k}, \qquad g_{k+1} = \frac{\partial}{\partial \theta} R(\theta, D_{k+1})\Big|_{\theta=\hat{\theta}_k} \tag{5}$$

both under cross-entropy loss. If $\langle g_k^*, g_{k+1} \rangle < 0$, apply the magnitude-aware projection (Eq. 3), and update with $\tilde{g}$ in place of the raw joint gradient. Since this projection depends only on the gradient vectors, not on how the loss producing them was defined, it transfers unchanged from the regression to the classification setting. Its effect is a hard guarantee: $\langle \tilde{g}, g_k^* \rangle \geq 0$, so no single step can increase the loss measured on the memory buffer – directly addressing MUDVI's residual negative BWT.

3.3 Addition 2 – Confidence-Based Relationship-Shift Detection

The regression method's residual $r_t = y_t - f(x_t; \theta)$ has no classification equivalent, since subtraction is not meaningful between class labels. We replace it with a scalar loss or confidence signal computed under a **frozen reference model** $\hat{\theta}_{D_1}$, trained once on the first subject/segment and never updated:

$$c_t = \ell_t(\hat{\theta}_{D_1}) = -\log \left[f(x_t; \hat{\theta}_{D_1}) \right]_{y_t} \tag{6}$$

or, when ground truth is unavailable at inference time, the model's own top confidence:

$$c_t = \max_k \left[f(x_t; \hat{\theta}_{D_1}) \right]_k \tag{7}$$

The resulting stream $c_1, c_2, \ldots$ is scalar and continuous, so the original NOT/GLR change-point test can be applied directly:

$$\Re^c_{(s,e]}(c) = 2 \log \left[\frac{\sup_{\eta_1, \eta_2} L(c_{s+1}, \ldots, c_c; \eta_1)\, L(c_{c+1}, \ldots, c_e; \eta_2)}{\sup_{\eta} L(c_{s+1}, \ldots, c_e; \eta)} \right] \tag{8}$$

A detected shift in c_t indicates the mapping from EEG signal to class has changed – e.g. electrode drift, fatigue, or a genuine new subject – even when MUDVI's existing MMD-based feature-distribution detector shows no shift. This detector is run **alongside** MUDVI's original kernel-MMD detector rather than replacing it: a subject shift is flagged if either detector fires, since both existing detection accuracies (72.5%–83.6%) leave meaningful room for missed detections, and the two detectors are sensitive to different failure modes (marginal input shift vs. relationship shift).

3.4 What Remains Unchanged

- **Volume/informativeness-aware memory update** (Eqs. 1–2): the source regression framework does not model imbalanced arrival volumes across segments, so MUDVI's own solution is retained as-is.
- **Cluster-based buffer architecture**: both additions plug into the existing per-subject clusters without requiring restructuring.

4 Combined Algorithm

Algorithm 1 GP-MUDVI: Volume-Aware Buffer with Gradient Projection and Dual Shift Detection

Require: Streaming EEG data D_t; memory buffer M; frozen reference model $\hat{\theta}_{D_1}$; number of detected subjects L_t

if MMD shift detected **or** confidence shift detected via c_t **then**
    $L_t \leftarrow L_{t-1} + 1$; create new cluster $M_{L_t} = \emptyset$
end if
if M has free space **then**
    $M_{L_t} \leftarrow M_{L_t} \cup D_t$
else
    compute P_t^{in} (Eq. 2)
    if D_t selected into memory **then**
        determine cluster C via $\mathcal{A}_C$ (Eq. 1)
        replace lowest-importance sample in M_C with D_t
    end if
end if
// Constrained joint training step
sample batch from M
$g_k^* \leftarrow \nabla_\theta R(\theta, M_{\text{sample}})$ ▷ cross-entropy
$g_{k+1} \leftarrow \nabla_\theta R(\theta, D_{\text{incoming}})$ ▷ cross-entropy
if $\langle g_k^*, g_{k+1} \rangle < 0$ **then**
    $\tilde{g} \leftarrow$ project (Eq. 3) smaller-norm gradient onto the larger
else
    $\tilde{g} \leftarrow g_k^* + g_{k+1}$
end if
update θ using $\tilde{g}$
// Update relationship-shift signal
compute c_t (Eq. 6 or 7) under $\hat{\theta}_{D_1}$
append c_t to change-point monitoring stream
return updated M, updated θ

5 Expected Effect on Evaluation Metrics

Following the evaluation protocol of Duan et al. (accuracy after sequential learning ends, and BWT), the two additions target specific weaknesses visible in their reported results:

- **Backward transfer (BWT).** Since gradient projection enforces $\langle \tilde{g}, g_k^* \rangle \geq 0$ at every step, forgetting caused purely by conflicting gradient directions should be reduced, directly targeting the negative BWT values reported for MUDVI.
- **Subject-shift detection accuracy.** Running the confidence-based detector alongside the MMD detector should raise the fraction of correctly detected shifts above MUDVI's reported 72.5%–83.6% range, since the two detectors are sensitive to different shift types (marginal vs. relationship).
- **Subject-agnostic vs. subject-aware gap.** MUDVI shows a consistent accuracy drop when subject identity is unknown. Since the confidence-based detector adds an independent signal for shift detection, this gap is expected to narrow when both detectors are combined.

6 Discussion

The two additions were chosen specifically because they operate at a level of abstraction – gradient vectors and scalar loss/confidence streams – that does not depend on whether the underlying task is regression or classification. This is why the projection rule (Section 3.2) transfers with no modification, while the shift-detection mechanism (Section 3.3) required a genuine reformulation: a numeric residual has no classification analogue, but a frozen-reference-model loss or confidence stream serves the same functional role of exposing relationship drift independent of input distribution drift.

The volume-imbalance handling in MUDVI was deliberately left untouched, since it solves a problem specific to the EEG setting – subjects producing highly unequal data volumes – which has no counterpart in the segment-based regression framework this work draws from.

References

- T. Duan, Z. Wang, F. Li, G. Doretto, D. Adjeroh, Y. Yin, C. Tao, *Online Continual Decoding of Streaming EEG Signal with a Balanced and Informative Memory Buffer*.
- W. Boulabiar, *Proposed Solution: A Continual Learning Framework for Glucose Insulin Dynamics, with Application to the OhioT1DM Dataset*.